

UGANDA MULTI-SCHOOL ERP

Complete System Architecture & Design Report

(Updated with Pages 61-66: CRUD Architecture)



PAGE 1 — SYSTEM OVERVIEW

A centralized SaaS ERP platform designed to manage multiple schools in Uganda with complete automation of academic, administrative, financial, and communication processes.



PAGE 2 — SYSTEM PURPOSE

TO REPLACE MANUAL SCHOOL MANAGEMENT WITH A FULLY DIGITAL SYSTEM THAT ENSURES:

- Accuracy
- Transparency
- Automation
- Accountability
- Scalability



PAGE 3 — CORE DESIGN PRINCIPLES

- Role-Based Access Control (RBAC)
- Multi-school isolation (multi-tenant SaaS)
- Workflow-driven operations
- Audit-first system design



PAGE 4 — SCHOOL STRUCTURE (UGANDA MODEL)

NURSERY:

- Baby Class
- Middle Class
- Top Class

PRIMARY:

P1–P7

SECONDARY:

S1–S6



PAGE 5 — ROLE SYSTEM OVERVIEW (FINAL CLEAN)

SYSTEM USERS ONLY:

- Super Admin
- School Admin
- Academic Coordinator
- Admission Staff
- Fee Officer
- Teacher
- Parent
- Auditor



PAGE 6 — SYSTEM ENTITIES (NOT ROLES)

- Students (records only)
- Classes
- Subjects
- Exams
- Fees
- Attendance
- Library records
- Sports records



PAGE 7 — SUPER ADMIN ROLE (WHY + FUNCTION + USE)

WHY THIS ROLE EXISTS

To control entire SaaS platform across all schools.

FUNCTIONS

- Manage schools
- Monitor system usage
- Global analytics

SUB-FUNCTIONS

- ✓ Create school accounts
- ✓ Manage subscriptions
- ✓ View all school reports
- ✓ System configuration
- ✓ Platform backup

USE

Ensures centralized governance of all schools.



PAGE 8 — SCHOOL ADMIN ROLE (WHY + FUNCTION + USE)

WHY

Acts as principal-level authority for each school.

FUNCTIONS

- Manage academics
- Approve admissions
- Control fees
- Manage staff structure

SUB-FUNCTIONS

- ✓ Class & section creation
- ✓ Subject approval
- ✓ Final admission approval
- ✓ Scholarship approval
- ✓ Fee structure approval
- ✓ Teacher assignment

USE

Controls full school operations.



PAGE 9 — ACADEMIC COORDINATOR ROLE

WHY

Handles academic structure and planning.

FUNCTIONS

- Curriculum control
- Timetable planning

SUB-FUNCTIONS

- ✓ Subject mapping
- ✓ Timetable generation
- ✓ Exam scheduling
- ✓ Academic calendar setup

USE

Ensures structured academic delivery.



PAGE 10 — ADMISSION STAFF ROLE

WHY

Handles admission workload efficiently.

FUNCTIONS

- Application processing
- Verification
- Shortlisting

SUB-FUNCTIONS

- ✓ Application intake
- ✓ Document verification
- ✓ Eligibility check
- ✓ Candidate scoring
- ✓ Shortlisting
- ✓ Scholarship request initiation

USE

Speeds up admission pipeline.



PAGE 11 — FEE OFFICER ROLE

WHY

Manages financial system.

FUNCTIONS

- Fee management
- Invoice generation

SUB-FUNCTIONS

- ✓ Fee structure setup
- ✓ Invoice generation
- ✓ Payment tracking
- ✓ Receipt generation
- ✓ Due tracking
- ✓ Scholarship adjustment handling

USE

Ensures financial control and accuracy.



PAGE 12 — TEACHER ROLE

WHY

Academic execution layer.

FUNCTIONS

- Teaching data entry
- Student evaluation

SUB-FUNCTIONS

- ✓ Attendance marking
- ✓ Marks entry
- ✓ Assignments
- ✓ Exam evaluation
- ✓ Student feedback

USE

Captures academic performance data.



PAGE 13 — PARENT ROLE

WHY

Communication bridge between school and home.

FUNCTIONS

- Student monitoring
- Fee payment
- Notifications

SUB-FUNCTIONS

- ✓ Attendance view
- ✓ Result view
- ✓ Fee tracking
- ✓ Payments
- ✓ Notifications
- ✓ Report download

USE

Improves transparency and engagement.



PAGE 14 — AUDITOR ROLE

WHY

Ensures system integrity.

FUNCTIONS

- Audit tracking
- Fraud detection

SUB-FUNCTIONS

- ✓ Login logs
- ✓ Financial audit
- ✓ Academic change logs
- ✓ Role activity logs

USE

Ensures accountability.



PAGE 15 — STUDENT MODULE (ENTITY ONLY)

FUNCTIONS

- ✓ Academic tracking
- ✓ Attendance records
- ✓ Fee mapping
- ✓ Profile data



PAGE 16 — ADMISSION MODULE OVERVIEW

WORKFLOW:

**Application → Verification → Shortlist → Scholarship → Approval
→ Student Creation**



PAGE 17 — APPLICATION MODULE

- ✓ Online form
- ✓ Document upload
- ✓ Class selection



PAGE 18 — VERIFICATION MODULE

- ✓ Age validation
- ✓ Document check
- ✓ Eligibility rules



PAGE 19 — SHORTLISTING MODULE

- ✓ Ranking system
- ✓ Evaluation scoring
- ✓ Candidate filtering



WHY

- To provide financial support control system

TYPES

- Merit-based
- Need-based
- Admission discount
- Emergency support

SUBMODULES

- ✓ Request system
- ✓ Evaluation engine
- ✓ Approval workflow
- ✓ Fee adjustment engine
- ✓ Lock system
- ✓ Audit tracking



PAGE 21 — SCHOLARSHIP WORKFLOW DIAGRAM

- Request Created
- Admission Staff Review
- Fee Officer Evaluation
- School Admin Approval
- Fee Adjustment Applied
- Audit Log Stored



PAGE 22 — FINAL ADMISSION MODULE

- ✓ Student creation
- ✓ Class assignment
- ✓ Section assignment



PAGE 23 — STUDENT MANAGEMENT MODULE

- ✓ Promotion
- ✓ Transfer
- ✓ Profile updates



PAGE 24 — ACADEMIC MODULE

- ✓ Class setup
- ✓ Subject mapping
- ✓ Curriculum design
- ✓ Timetable generation



PAGE 25 — ATTENDANCE MODULE

- ✓ Student attendance
- ✓ Staff attendance
- ✓ Reports



PAGE 26 — EXAM MODULE

- ✓ Exam creation
- ✓ Marks entry
- ✓ Result generation
- ✓ Lock system



PAGE 27 — RESULT MODULE

- ✓ Report cards
- ✓ Grades
- ✓ Performance analytics



PAGE 28 — FEE MODULE

- ✓ Fee structure
- ✓ Invoice
- ✓ Payments
- ✓ Receipts



PAGE 29 — PAYMENT WORKFLOW

- Fee Setup
- Invoice Generated
- Parent Payment
- Verification
- Receipt Generated
- Ledger Updated



PAGE 30 — COMMUNICATION MODULE

- ✓ SMS alerts
- ✓ Email alerts
- ✓ Notifications



PAGE 31 — REPORTING MODULE

- ✓ Academic reports
- ✓ Fee reports
- ✓ Attendance reports
- ✓ Scholarship reports



PAGE 32 — AUDIT MODULE

- ✓ System logs
- ✓ Financial tracking
- ✓ Academic changes



PAGE 33 — SECURITY SYSTEM

- ✓ JWT authentication
- ✓ Role permissions
- ✓ School isolation



PAGE 34 — WORKFLOW ENGINE

Admission → Academic → Finance → Reporting



PAGE 35 — DATA RULE ENGINE

- ✓ No hard delete
- ✓ Soft delete only
- ✓ Locked records



PAGE 36 — MULTI-SCHOOL MODEL

- ✓ Each school isolated
- ✓ Central SaaS control



PAGE 37 — MOBILE MONEY INTEGRATION

- ✓ MTN Mobile Money
- ✓ Airtel Money



PAGE 38 — LIBRARY MODULE (NEW REQUEST)

- ✓ Book management
- ✓ Issue/return tracking
- ✓ Fine system
- ✓ Student/teacher access



PAGE 39 — SPORTS MODULE

- ✓ Team registration
- ✓ Event management
- ✓ Attendance tracking
- ✓ Performance records



PAGE 40 — ACTIVITIES MODULE

- ✓ School events
- ✓ Participation tracking
- ✓ Certificates



PAGE 41 — DASHBOARD SYSTEM

- ✓ Admin dashboard
- ✓ Teacher dashboard
- ✓ Parent dashboard



PAGE 42 — NOTIFICATION ENGINE

- ✓ SMS
- ✓ Email
- ✓ Event triggers



PAGE 43 — BACKUP SYSTEM

- ✓ Daily backups
- ✓ School-wise restore



PAGE 44 — ERROR HANDLING

- ✓ Validation system
- ✓ Exception logs



PAGE 45 — SCALABILITY DESIGN

- ✓ Microservice-ready
- ✓ Horizontal scaling



PAGE 46 — API STRUCTURE

- ✓ REST APIs per module
- ✓ Role-based endpoints



PAGE 47 — DATABASE OVERVIEW

- ✓ Users
- ✓ Students
- ✓ Fees
- ✓ Exams



PAGE 48 — ADMISSION DATABASE FLOW

- ✓ Applications
- ✓ Verification
- ✓ Approval



PAGE 49 — FEE DATABASE FLOW

- ✓ Invoices
- ✓ Payments
- ✓ Scholarship mapping



PAGE 50 — AUDIT DATABASE

- ✓ Logs
- ✓ Change history



PAGE 51 — ROLE PERMISSION MATRIX

- ✓ Create
- ✓ Read
- ✓ Update
- ✓ Approve



PAGE 52 — BUSINESS RULES

- ✓ No fee editing after payment
- ✓ No result editing after publish



PAGE 53 — INTEGRATION SYSTEM

- ✓ SMS gateway
- ✓ Email system
- ✓ Mobile money APIs



PAGE 54 — USER FLOW

Login → Dashboard → Module → Action → Audit



PAGE 55 — PERFORMANCE TRACKING

- ✓ Students
- ✓ Teachers
- ✓ Schools



PAGE 56 — SYSTEM BENEFITS

- ✓ Automation
- ✓ Transparency
- ✓ Scalability



PAGE 57 — FUTURE ENHANCEMENTS

- ✓ AI analytics
- ✓ Mobile apps
- ✓ Biometric attendance



PAGE 58 — DEPLOYMENT PLAN

- ✓ Database setup
- ✓ Backend build
- ✓ Testing
- ✓ Deployment



PAGE 59 — CLIENT VALUE

- ✓ One system for all operations
- ✓ Real-time control
- ✓ Financial transparency



PAGE 60 — FINAL CONCLUSION

THIS ERP IS A COMPLETE UGANDA MULTI-SCHOOL SAAS PLATFORM WITH:

- ✓ Full role-based functionality
- ✓ Complete module & submodule design
- ✓ Workflow-driven architecture
- ✓ Library + Sports + Activities integration
- ✓ Strong audit & security



PAGE 61 — WHERE DATA IS MANAGED (CRITICAL ERP DESIGN RULE)

In this ERP system, data is NOT managed randomly across roles.

Instead, all core records are managed inside dedicated centralized modules with controlled access and workflows.

THE TWO MOST IMPORTANT DATA ENTITIES IN THE SYSTEM ARE:

- 🎓 Students (core academic entity)



Staff (teachers + administrative + operational school staff)

PAGE 62 — STUDENT MANAGEMENT (CRUD ARCHITECTURE)

MODULE:

 Student Management Module

This module is the single source of truth for all student records.

WHY THIS MODULE EXISTS

STUDENT DATA IS CONNECTED TO:

- Admission
- Attendance
- Exams
- Fees
- Scholarships
- Promotion
- Reports

So it must be centralized to avoid duplication and errors.

CRUD OPERATIONS (CONTROLLED MODEL)

- ✓ CREATE (Student Creation)

Students are NOT created manually like normal data.

THEY ARE CREATED ONLY WHEN:

Admission Approved → School Admin Final Approval → Student Record Created

CREATED BY:

- Admission Staff (draft record)
- School Admin (final activation)
- ✓ READ (View Student Data)

ACCESS DEPENDS ON ROLE:

 **Teacher → Only assigned class students**



School Admin → Full school access



Parent → Only linked student



Auditor → Full system access (read-only)

- ✓ UPDATE (Modify Student Data)

ALLOWED ONLY IN CONTROLLED CONDITIONS:



ALLOWED ROLES:

- Admission Staff (before approval)
- School Admin (after approval with audit log)



ALLOWED UPDATES:

- Name correction
- Class change
- Section change
- Contact update
- Status update
- ✗ DELETE (STRICT RULE)
- No permanent deletion allowed

SYSTEM USES:

- ✓ Soft delete
- ✓ Archive status
- ✓ Transfer status
- 📌 Only School Admin / Super Admin can deactivate records



STUDENT SUBMODULES

1. Student Profile Module

- Personal details
- Guardian details
- Admission info

2. Class Assignment Module

- Assign class (N/P/S levels)

- Section allocation

3. Promotion Module

- Move student to next class annually

4. Transfer Module

- Inter-class / inter-school movement

5. Status Control Module

- Active
- Inactive
- Graduated
- Transferred

PURPOSE OF STUDENT MODULE

- ✓ Ensures centralized academic identity
- ✓ Prevents duplicate records
- ✓ Links all academic + financial operations
- ✓ Maintains full lifecycle tracking

PAGE 63 — STAFF MANAGEMENT (CRUD ARCHITECTURE)

MODULE:

 Staff Management Module (Teachers + School Operational Staff)

WHY THIS MODULE EXISTS

STAFF DATA IS REQUIRED FOR:

- Teaching assignments
- Academic coordination
- Library operations
- Sports activities
- Fee processing support
- System accountability

CRUD OPERATIONS

- ✓ CREATE (Staff Registration)

CREATED ONLY BY:

- School Admin

STAFF TYPES:

- Teacher
- Librarian
- Sports Coach
- Academic Staff
- Fee Officer
- ✓ READ (View Staff Data)

ACCESS:

School Admin → Full access

Academic Coordinator → Teacher-only data

Auditor → Read-only full access

- ✓ UPDATE (Modify Staff Data)

ALLOWED ONLY BY:

- School Admin

UPDATES INCLUDE:

- Role assignment
- Subject assignment
- Department change
- Contact updates
-  DELETE (STRICT CONTROL)
- No permanent deletion allowed

STAFF STATUS IS ONLY:

- ✓ Active
- ✓ Inactive
- ✓ Transferred
- ✓ Resigned

STAFF SUBMODULES

1. Staff Profile Module

- Personal information
- Qualification details
- Contact information

2. Role Assignment Module

- Teacher role
- Librarian role
- Sports role
- Admin role

3. Academic Allocation Module

- Subject assignment
- Class assignment
- Timetable linking

4. Attendance Module

- Daily attendance tracking
- Monthly summaries

5. Performance Tracking Module

- Evaluation records
- Work performance logs

PURPOSE OF STAFF MODULE

- ✓ Controls all human resources in school
- ✓ Links teachers to academic system
- ✓ Supports operational activities (library, sports, etc.)
- ✓ Ensures accountability and tracking



PAGE 64 — IMPORTANT ERP DESIGN RULE (CRITICAL UNDERSTANDING)



ERP RULE: “CRUD IS NEVER DIRECT”

IN THIS SYSTEM:

CRUD operations are NOT freely accessible actions.

INSTEAD:

- ✓ Every CRUD action is controlled
- ✓ Every change goes through workflow
- ✓ Every change is logged in audit system



CONTROLLED DATA FLOW MODEL

Request → Role Validation → Approval → Execution → Audit Log → Lock



EXAMPLE

- ✗ WRONG SYSTEM

Teacher directly edits marks or student profile

- ✓ YOUR ERP SYSTEM

Teacher submits data → Admin verifies → System locks record



PAGE 65 — WHY THIS DESIGN IS IMPORTANT

THIS CONTROLLED CRUD SYSTEM ENSURES:

- ✓ DATA ACCURACY
- No unauthorized edits
- ✓ TRANSPARENCY
- Every action is tracked
- ✓ SECURITY
- No role can bypass system
- ✓ REAL SCHOOL SIMULATION
- Matches real Uganda school operations
- ✓ SCALABILITY
- Can support thousands of schools safely

PAGE 66 — FINAL SUMMARY OF DATA CONTROL SYSTEM

YOUR ERP SYSTEM IS BUILT ON:

CORE IDEA:

“Data is created by one role, validated by another, approved by another, and locked by the system.”

• FINAL STRUCTURE

Entity	Managed By
Students	Admission Staff + School Admin
Staff	School Admin
Academic Data	Teachers + Academic Coordinator
Fees	Fee Officer
Results	Teachers + Admin
Audit Logs	System (Automatic)